

Cooperative Post-Harvest Loss Tracker

Project Description — NISR 2026 Big Data Hackathon (Track 1: Agricultural Productivity)

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The Problem

After farmers harvest their crops in Rwanda, a significant portion of that harvest is lost before it ever reaches a market or a plate. Crops get stolen, eaten by pests or birds, damaged during harvesting, lost in transport, spoiled in poor storage, or wasted during processing and packaging. This is called "post-harvest loss," and it directly hurts farmers' income and Rwanda's food security — it is a recognized priority area under Rwanda's national development plan (NST2).

Most agricultural cooperatives currently have no easy way to track how much they are losing, why they are losing it, or how their losses compare to other cooperatives nearby. Without this information, it is difficult to know where to focus efforts to reduce losses.

Our Solution

We are building a simple digital tool that helps agricultural cooperatives track their harvest losses and understand where those losses come from — then compares their numbers to real national data so they know whether they are doing better or worse than average. The tool works as follows:

- A cooperative logs their harvest — how much they grew for each crop, in which district, during which season.
- They also log their losses, broken down by cause: theft, pests, birds, harvesting damage, transport, storage, processing, packaging, or loss at the point of sale.
- The tool compares their loss numbers to real government survey data (from Rwanda's National Institute of Statistics) for their district and crop, showing whether their loss rate is normal or higher than expected.
- If a cooperative's loss rate is high, the tool identifies the most likely cause based on patterns in the national data — for example, flagging that storage-related losses are unusually high in their district, pointing to where investment is needed.
- A predictive feature uses a machine learning model trained on real historical data to estimate the expected loss risk for a cooperative's crop and season before the season happens — functioning as an early warning system.

Who It Helps

- **Cooperatives** — to understand and reduce their losses, protecting farmer income.
- **Government and development partners** — to see, at a glance, which districts and crops have the highest loss rates, guiding where to focus support programs.
- **Civil society organizations** — to use real, localized data when advocating for better storage infrastructure or farmer training.

The Data Behind It

The project uses real, official data from Rwanda's National Institute of Statistics (NISR) — specifically the Seasonal Agricultural Survey, which already asks farmers directly about their harvest losses and the reasons behind them. This means the tool is not guessing — it is built on real government data, including a detailed, cause-attributed breakdown of losses (theft, pests, birds, harvesting, transport, storage, processing, packaging, and sale).

The Technology

- Raw survey data is cleaned and processed using Python.
- A machine learning model (scikit-learn) predicts expected loss risk for a given district, crop, and season.
- The prediction is served through a lightweight API (FastAPI) that the web app calls in real time.
- The web app itself — where cooperatives log data and view their dashboard — is built with a modern web stack, featuring clear charts, simple forms, and an easy-to-read comparison view.

Why It Matters

This is not just a data dashboard — it turns raw government statistics into something a cooperative leader can act on, such as "we need better storage" or "we are losing too much during transport, let's address that." It is a focused tool aimed at a real, measurable problem that affects farmer livelihoods across the country.